UPLINK INTEGRATED SOLUTIONS INC.
System and Network Department

EMS Gateway API Guide
Technical Document

Live and historical data from air quality, seismic and water level stations over the gateway's REST API

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1. Purpose and scope

This document describes how to read data from the EMS Gateway through its REST API. Live and historical readings from the gateway's air quality, seismic and water level stations, plus the gateway's own health and logs. Every endpoint is a `GET` that returns JSON; nothing in the API changes data.

2. Quick start

1. **Create a key** —On the dashboard, open ⚙ **Settings › API** (super admins). Add a key with a label for who uses it, and copy it right away: the gateway keeps only a hash, so the key is shown once.
2. **Allow your network** —On the same page, add the caller's IP or network (e.g. `192.168.1.0/24`) to **Allowed networks**. With an empty list, every request is refused, the health check included.
3. **Call it** —Send the key in the `X-API-Key` header. New keys and networks take effect within 5 minutes, without a restart.

CURL

```
curl -s -H "X-API-Key: YOUR_API_KEY" \
  http://GATEWAY_IP/api/water-level/stations/latest
```

PYTHON

```
import requests

BASE = "http://GATEWAY_IP/api"
HEADERS = {"X-API-Key": "YOUR_API_KEY"}

r = requests.get(f"{BASE}/water-level/stations/latest", headers=HEADERS, timeout=10)
r.raise_for_status()
for s in r.json()["stations"]:
    print(s["station_mn"], s["status"], s["water_level_m"], "m")
```

JAVASCRIPT

```
const BASE = "http://GATEWAY_IP/api";

const res = await fetch(`${BASE}/water-level/stations/latest`, {
  headers: { "X-API-Key": "YOUR_API_KEY" },
});
if (!res.ok) throw new Error(`${res.status}: ${(await res.json()).detail}`);
const { stations } = await res.json();
stations.forEach(s => console.log(s.station_mn, s.status, s.water_level_m));
```

3. Access: address, keys and limits

Item	Value
Base URL	`http://GATEWAY_IP/api` on port 80, or `http://GATEWAY_IP:8443/api` (API-only port). Both go through nginx to the API service.
Authentication	Header `X-API-Key: <key>` on every endpoint except `/api/system/status`.
IP allowlist	Checked before the key. The caller's address must be inside one of the **Allowed networks** (single IP or CIDR).

Rate limit	Per key, a sliding 60-second window. Default 9,999 requests per 60 s (`API_RATE_LIMIT_REQUESTS` , `API_RATE_LIMIT_WINDOW_SEC`). Over the limit: `429` with a `Retry-After` header.
Logging	Every request is recorded on the dashboard's **API Logs** page with the key's label and only its last 4 characters.
Interactive docs	`/api/docs` (Swagger UI, try requests in the browser), `/api/redoc` , and the OpenAPI schema at `/api/openapi.json` for code generators.

> Treat a key like a password: one key per client system, so you can see each one in the API Logs and delete one without breaking the others.

4. Times and station status

Every time the API returns is Philippine time with its offset, for example `2026-10-26T15:42:10.123+08:00` . Any ISO 8601 parser reads it correctly.

Station type	status = online	idle	offline
Air quality	last reading under 15 min ago	—	otherwise
Seismic	last reading under 5 min ago	—	otherwise
Water level	within its reporting interval + 2 min	within 2 × interval + 2 min	otherwise

The top-level `timestamp` is the time of the request, except on `/air-quality/stations/latest` , where it is the newest reading across all stations.

5. Errors

Errors return a JSON body with a `detail` field.

Code	detail	What to do
403	`Unauthorized request: IP address not allowed`	Add the caller's IP or network under Allowed networks.
403	`Unauthorized request: Invalid API Token` or `Not authenticated`	Send a valid key in `X-API-Key` . A key deleted on the dashboard stops working within 5 min.
404	`Station Identifier not found in database records.`	Check the station code. Codes are case-sensitive.
422	a list of validation problems	A query parameter is out of range, e.g. `hours=48` on seismic history (max 24).
429	`Rate limit exceeded: max N requests per 60s per API token.`	Wait the number of seconds in `Retry-After` .
500	`Internal Server Error`	The gateway couldn't read its database. Check `/api/system/status` and the Logs page.

6. Air quality endpoints

Readings from the HJ 212 air quality stations. A measurement the station has never sent reads as `0`, not `null`.

6.1 GET /api/air-quality/stations/latest

Latest reading for every registered air quality station: location, status, weather and pollutants.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/air-quality/stations/latest
```

EXAMPLE RESPONSE

```
{
  "timestamp": "2026-10-26T15:40:00.000+08:00",
  "total_stations": 1,
  "stations": [
    {
      "station_mn": "4101025U122041",
      "friendly_name": "AQM001",
      "location": { "latitude": 14.5995, "longitude": 120.9842 },
      "status": "online",
      "last_update": "2026-10-26T15:40:00.000+08:00",
      "weather": { "temperature": 29.4, "humidity": 68.2, "pressure": 101.3, "rain": 0.0,
                   "wind_speed": 2.1, "wind_direction": 180.0, "noise": 62.5 },
      "pollutants": { "pm2_5": 18.3, "pm10": 32.7, "tsp": 45.1, "co": 0.6, "so2": 5.2,
                      "no2": 21.4, "o3": 38.9, "pb": 0.02, "pb_temp": 27.8,
                      "estimated": false, "estimated_source": null }
    }
  ]
}
```

`pollutants.estimated: true` means PM2.5, PM10, TSP and NO_2 were filled in from station `estimated_source` because this station has no live reading of its own. Treat those four values as estimates.

6.2 GET /api/air-quality/stations/{station_mn}/latest

Latest reading for one station, by its monitoring number. Same station object as above, under `station`; `timestamp` is that reading's time.

Parameter	In	Description
`station_mn`	path	Monitoring number, e.g. `4101025U122041`. Unknown → `404`.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/air-quality/stations/4101025U122041/latest
```

6.3 GET /api/air-quality/analytics/1d

One row per station with each value averaged over the last 24 hours (rounded to 2 decimals). Stations with no readings in that time are left out.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/air-quality/analytics/1d
```

EXAMPLE RESPONSE

```
{
  "range": "24_hours_aggregated_average",
  "timestamp": "2026-10-26T15:40:00.000+08:00",
  "results": [
    { "station_mn": "4101025U122041", "station_name": "AQM001",
      "temperature": 28.9, "humidity": 70.1, "air_pressure": 101.2, "rain": 0.4,
      "wind_speed": 2.3, "wind_direction": 175.0, "noise": 60.8,
      "pm25": 19.1, "pm10": 33.4, "tsp": 46.0, "carbon_monoxide": 0.58,
      "sulfur_dioxide": 5.0, "nitrogen_dioxide": 20.7, "ozone": 37.2,
      "lead": 0.02, "lead_temperature": 27.5 }
  ]
}
```


> The analytics endpoints use the database column names (`pm25` , `carbon_monoxide` , `air_pressure` …), not the short names of the live endpoints (`pm2_5` , `co` , `pressure`). The field dictionary maps them.

6.4 GET /api/air-quality/analytics/7d and /api/air-quality/analytics/30d

Daily averages per station for the last 7 or 30 days: one row per station per day, newest day first, each with a `summary_date` (`YYYY-MM-DD`). `range` is `7_days_daily_averages` or `30_days_daily_averages` .

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/air-quality/analytics/7d
```

6.5 GET /api/air-quality/stations

The station registry without readings: `station_mn` , `station_name` , `latitude` , `longitude` , `updated_at` , plus `total_registered` . Use it to discover station codes.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/air-quality/stations
```

7. Seismic endpoints

Three-component readings from the seismic stations, received over MQTT or SMS. `x` and `y` are the horizontal axes and `z` is vertical; values are passed through as the station sends them.

7.1 GET /api/seismic/stations/latest

Latest reading for every station that has ever reported: location, status, acceleration, velocity, displacement, graph values, PGA and PEIS.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/seismic/stations/latest
```

EXAMPLE RESPONSE

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```
{
  "timestamp": "2026-10-26T15:42:10.512+08:00",
  "total_stations": 1,
  "stations": [
    {
      "station_id": "STN-001",
      "friendly_name": "Pacific Ridge Station",
      "location": { "latitude": 14.5995, "longitude": 120.9842, "elevation_m": 12.5 },
      "status": "online",
      "last_update": "2026-10-26T15:42:09.000+08:00",
      "acceleration": { "x": 0.0021, "y": -0.0013, "z": 0.0042 },
      "velocity":     { "x": 0.0008, "y": -0.0004, "z": 0.0011 },
      "displacement": { "x": 0.0002, "y": -0.0001, "z": 0.0003 },
      "graph": { "x": 0.0021, "y": -0.0013, "z": 0.0042, "scale": "1:100", "center": 0.0 },
      "pga": 0.031,
      "peis": 5
    }
  ]
}
```


7.2 GET /api/seismic/stations/{station_id}/latest

The same station object for one station, under `station`.

Parameter	In	Description
`station_id`	path	Station identifier, e.g. `STN-001`. Unknown → `404`.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/seismic/stations/STN-001/latest
```

7.3 GET /api/seismic/graph/latest

Only identity, status and the `graph` object for every station: a small payload for screens that draw a live trace. Poll it (e.g. once a second) and append each new `last_update` to your trace; skip a poll that returns the same `last_update`.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/seismic/graph/latest
```

EXAMPLE RESPONSE


```
{
  "timestamp": "2026-10-26T15:42:10.512+08:00",
  "total_stations": 1,
  "stations": [
    { "station_id": "STN-001", "friendly_name": "Pacific Ridge Station", "status": "online",
      "last_update": "2026-10-26T15:42:09.000+08:00",
      "graph": { "x": 0.0021, "y": -0.0013, "z": 0.0042, "scale": "1:100", "center": 0.0 } }
  ]
}
```


7.4 GET /api/seismic/stations/{station_id}/graph/latest

The graph object for one station. Unknown station → `404`.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/seismic/stations/STN-001/graph/latest
```

7.5 GET /api/seismic/stations/{station_id}/history

Every stored reading for one station within the window, oldest first: the data for a seismogram. It has no `graph` fields; use acceleration, velocity or displacement.

Parameter	In	Description
`station_id`	path	Station identifier. An unknown station returns an empty `readings` list, not 404.
`hours`	query	Look-back window, **1–24**, default 1.

```
curl -s -H "X-API-Key: YOUR_API_KEY" "http://GATEWAY_IP/api/seismic/stations/STN-001/history?hours=1"
```

EXAMPLE RESPONSE

```
{
  "station_id": "STN-001",
  "hours": 1,
  "readings": [
    { "time": "2026-10-26T15:00:01.000+08:00",
      "acc_x": 0.0021, "acc_y": -0.0013, "acc_z": 0.0042,
      "vel_x": 0.0008, "vel_y": -0.0004, "vel_z": 0.0011,
      "disp_x": 0.0002, "disp_y": -0.0001, "disp_z": 0.0003,
      "pga": 0.031, "peis": 5 }
  ]
}
```

> Section 8 shows how this data looks when drawn as a seismogram.

7.6 GET /api/seismic/events

Readings from all stations whose PEIS is at least `min_peis`, newest first. Each event has `time`, `station_id`, `station_name`, `latitude`, `longitude`, `pga`, `peis`; the response adds `min_peis`, `hours`, `total_events`.

Parameter	In	Description
`min_peis`	query	Lowest PEIS to include, **0 or more**, default 1.
`hours`	query	Look-back window, **1–168** (7 days), default 24.

```
curl -s -H "X-API-Key: YOUR_API_KEY" "http://GATEWAY_IP/api/seismic/events?min_peis=3&hours=72"
```

8. Seismic data as a seismogram

The traces below show what the seismic history data looks like: one trace per axis, sharing a time line.

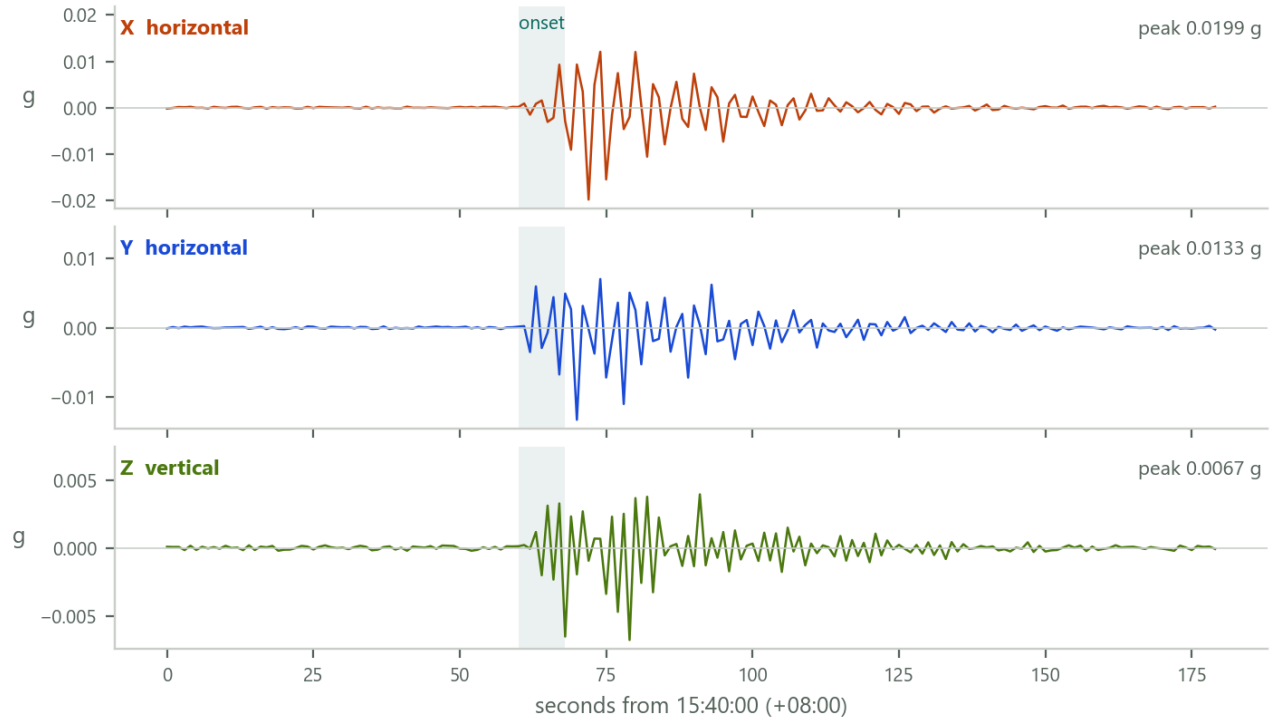


Figure 1. Three-axis seismogram drawn from the history endpoint (simulated event)

Figure 1 is a simulated 3-minute record at 1 reading per second, drawn from `acc_x`, `acc_y` and `acc_z`. The event starts at 60 s; peak PGA is 0.0199 g (PEIS 4).

8.1 Reading the traces

Quiet ground shows a thin band of noise around zero. An event shows as a burst on all three axes; the horizontal traces usually carry the larger swing. **PGA** is the peak ground acceleration of a reading and **PEIS** its intensity on the PHIVOLCS Earthquake Intensity Scale. Both are computed on the station. The bands below are the ones the gateway's station simulator uses (PGA in g); your station firmware may differ.

PEIS	Description	PGA
0	Not felt	0 – 0.001 g
1	Scarcely perceptible	0.001 – 0.002 g
2	Slightly felt	0.002 – 0.005 g
3	Weak	0.005 – 0.01 g
4	Moderately felt	0.01 – 0.02 g
5	Strong	0.02 – 0.04 g
6	Very strong	0.04 – 0.08 g
7	Damaging	0.08 – 0.15 g
8	Very damaging	0.15 – 0.25 g
9	Destructive	0.25 – 0.4 g
10	Very destructive	0.4 – 0.6 g

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11	Extremely destructive	0.6 – 0.85 g
12	Completely devastating	≥ 0.85 g

8.2 Drawing a live trace

For a scrolling live view, poll the graph endpoint and keep the last few minutes of points:

```
const BASE = "http://GATEWAY_IP/api";
const trace = [];             // [{t, x, y, z}]
let last = null;

async function poll() {
  const r = await fetch(`${BASE}/seismic/stations/STN-001/graph/latest`,
                        { headers: { "X-API-Key": "YOUR_API_KEY" } });
  const { station } = await r.json();
  if (station.last_update !== last) {          // a new reading arrived
    last = station.last_update;
    const g = station.graph;
    trace.push({ t: new Date(last), x: g.x - (g.center ?? 0), y: g.y - (g.center ?? 0), z: g.z - (g.center ?? 0) });
    while (trace.length > 300) trace.shift();  // keep ~5 min at 1 reading/s
    draw(trace);                               // your chart code
  }
}
setInterval(poll, 1000);
```

9. Water level endpoints

Ultrasonic water level stations reporting by SMS and over the LAN. The station measures the distance down to the water; the gateway computes `water_level_m = installation_height_m − distance_m`.

9.1 GET /api/water-level/stations/latest

Every enabled station with its newest reading, status and reporting interval.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/water-level/stations/latest
```

EXAMPLE RESPONSE

```
{
  "timestamp": "2026-10-26T15:42:10.512+08:00",
  "total_stations": 1,
  "stations": [
    {
      "station_mn": "WLS001",
      "station_name": "Sto. Nino Bridge",
      "location": { "name": "Marikina River, Sto. Nino Bridge", "latitude": 14.6335, "longitude": 121.096 },
      "status": "online",
      "report_interval_minutes": 3,
      "last_update": "2026-10-26T15:41:02.000+08:00",
      "water_level_m": 3.89,
      "distance_m": 1.61,
      "installation_height_m": 5.5,
      "battery_v": 8.75,
      "temperature_c": null
    }
  ]
}
```


`water_level_m` and `distance_m` are `null` when the sensor got no echo, and `water_level_m` also when no installation height is set. Treat `null` as "no reading", never as zero.

9.2 GET /api/water-level/stations/{station_mn}/latest

The same station object for one station, under `station`. Unknown or disabled station → `404`.

```
curl -s -H "X-API-Key: YOUR_API_KEY" http://GATEWAY_IP/api/water-level/stations/WLS001/latest
```

9.3 GET /api/water-level/stations/{station_mn}/history

The station's readings in the window, oldest first, at most 5,000. Each reading has `recorded_at`, `water_level_m`, `distance_m`, `battery_v`, `temperature_c`; the response adds `hours` and `total_readings`.

Parameter	In	Description
`station_mn`	path	Station code, e.g. `WLS001`. Unknown → `404`.
`hours`	query	Look-back window, **1–720** (30 days), default 24.

```
curl -s -H "X-API-Key: YOUR_API_KEY" "http://GATEWAY_IP/api/water-level/stations/WLS001/history?hours=168"
```

10. System endpoints

10.1 GET /api/system/status no key

Health check for uptime monitors: whether each database connection pool is ready. No key and no rate limit, but the caller must still be on the allowlist.

```
curl -s http://GATEWAY_IP/api/system/status
{
  "status": "operational",
  "timestamp": "2026-10-26T15:42:10.512+08:00",
  "subsystems": { "air_quality_db_pool": "initialized", "seismic_db_pool": "initialized",
                  "api_db_pool": "initialized", "water_level_db_pool": "initialized" }
}
```

10.2 GET /api/system/logs

The gateway services' log records, newest first: `created_at`, `service`, `level`, `logger_name`, `thread_name`, `message`, plus `total`.

Parameter	In	Description
`service`	query	One service, e.g. `air_quality_ingest`, `seismic_mqtt`, `api_server`, `water_level_gsm`.
`level`	query	`INFO`, `WARNING`, `ERROR` or `CRITICAL`.
`hours`	query	**1–168**, default 24.
`limit`	query	**1–1000**, default 200.

```
curl -s -H "X-API-Key: YOUR_API_KEY" "http://GATEWAY_IP/api/system/logs?service=seismic_mqtt&level=ERROR&hours=6"
```

11. Field dictionary

Values are stored as the stations send them. Units marked *typical* are the usual HJ 212 or sensor units; confirm against your station's configuration.

11.1 Air quality fields

Live field	Analytics field	Meaning	Unit
`weather.temperature`	`temperature`	Air temperature	°C
`weather.humidity`	`humidity`	Relative humidity	%
`weather.pressure`	`air_pressure`	Air pressure	kPa (typical)
`weather.rain`	`rain`	Rainfall	mm (typical)
`weather.wind_speed`	`wind_speed`	Wind speed	m/s
`weather.wind_direction`	`wind_direction`	Direction the wind comes from	° from north
`weather.noise`	`noise`	Noise level	dB(A)
`pollutants.pm2_5`	`pm25`	Fine particulate matter	µg/m³
`pollutants.pm10`	`pm10`	Coarse particulate matter	µg/m³
`pollutants.tsp`	`tsp`	Total suspended particles	µg/m³
`pollutants.co`	`carbon_monoxide`	Carbon monoxide	mg/m³ (typical)
`pollutants.so2`	`sulfur_dioxide`	Sulfur dioxide	µg/m³ (typical)

`pollutants.no2`	`nitrogen_dioxide`	Nitrogen dioxide	µg/m³ (typical)
`pollutants.o3`	`ozone`	Ozone	µg/m³ (typical)
`pollutants.pb`	`lead`	Lead (airborne)	µg/m³ (typical)
`pollutants.pb_temp`	`lead_temperature`	Lead sampler temperature	°C
`pollutants.estimated`	—	PM2.5, PM10, TSP, NO_2 filled from another station	true / false

11.2 Seismic fields

Field	Meaning	Unit
`acceleration.x/y/z`, `acc_x…`	Ground acceleration per axis (x, y horizontal; z vertical)	g (typical)
`velocity.x/y/z`, `vel_x…`	Ground velocity per axis	m/s (typical)
`displacement.x/y/z`, `disp_x…`	Ground displacement per axis	m (typical)
`pga`	Peak ground acceleration of the reading	g (typical)
`peis`	PHIVOLCS Earthquake Intensity Scale value from the station	integer
`graph.x/y/z`	Display values the station sends for a live trace	as sent
`graph.scale`	The station's display scale label, e.g. `1:100`	text
`graph.center`	Baseline of the trace; subtract it before drawing	as sent
`location.elevation_m`	Station elevation	m

11.3 Water level fields

Field	Meaning	Unit
`water_level_m`	Height of the water surface: installation height − distance	m
`distance_m`	Distance from the sensor down to the water (URM06, 0.20–10 m)	m
`installation_height_m`	Sensor height above the reference level, set on the dashboard	m
`battery_v`	Station supply voltage	V
`temperature_c`	Optional DS18B20 probe; `null` when not fitted	°C
`report_interval_minutes`	How often the station sends an SMS reading	min
`recorded_at` / `last_update`	When the reading was taken (SMS) or polled (LAN)	ISO 8601, +08:00

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